

FUNDAMENTALS OF COMPUTER NETWORK

* Computer Network

- Definition - A computer network is the interconnection of two or more computers to exchange data and to share resources with each other.
- Two computers are said to be interconnected if they exchange information.
- The computers connected in a network share files, folders, applications and resources like scanners, web-cams, printers etc.

* Need of Computer Network

1. For sharing the resources such as printers, among all the users.
2. For sharing of expensive software and database.
3. To facilities communication from one computer to the other.
4. To have exchange of data and information amongst the users, via the network.
5. For sharing of information over the geographically wide areas.
6. For connecting the computers between various buildings of an organization.
7. For educational purposes.

* Applications of Computer Network

1. Sharing of resources
2. Sharing of expensive software's and database

3. Communication from one computer to another.
4. Exchange of data and information among users via network.
5. Sharing of information over geographical wide areas.

* Components of a Computer Network

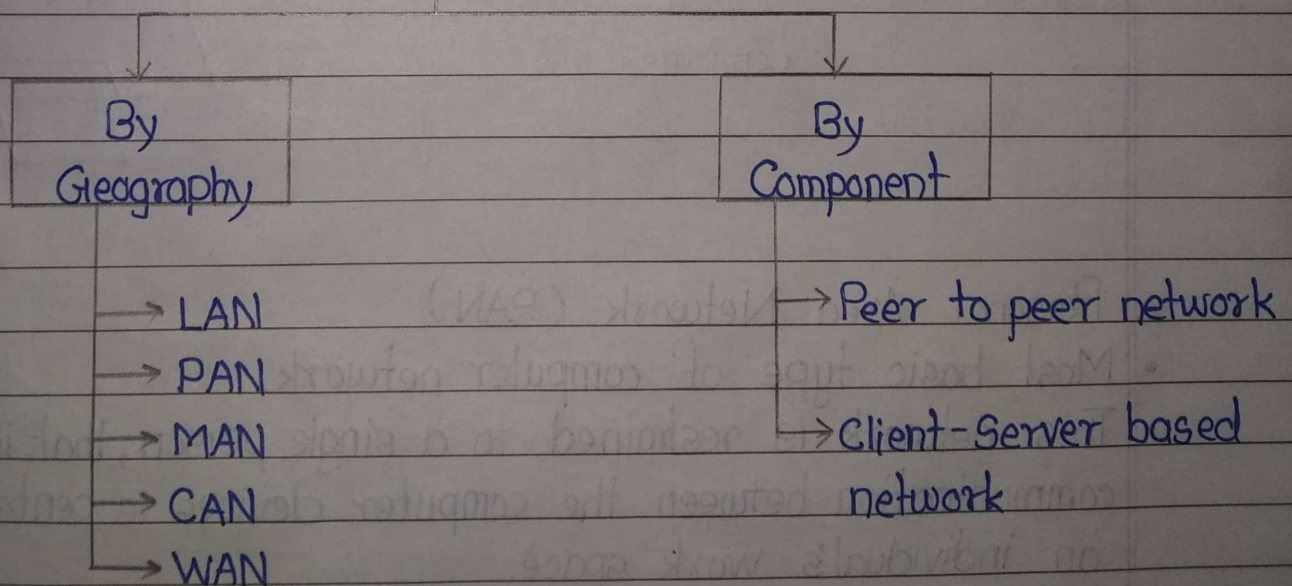
1. Two or more computers
2. Cables as links between the computers
3. A network interfacing card (NIC) on each computer.
4. Switches
5. A software called network operating system.

* Network Benefits

- File sharing
You can easily share data between different users, or access it remotely if you keep it on other connected devices.
- Resource Sharing
Using network connected peripheral devices like printers, scanners and copiers or sharing software between multiple users, saves money.
- Maintaining the Network
A well-maintained network ensures that your connectivity is optimized thus it is easy for users to access data and other resources without too much waiting.
- Backing up data
Network backup is essential to ensure the security of your business and all of its assets.

- **Sharing a single internet connection**
It is cost efficient and can help protect your systems if you properly secure the network.
- **Increasing storage capacity**
You can access files and multimedia, such as images and music, which you store remotely on other machines or network attached storage devices.
- **Improved communication**
Networking computers can also help you improve communication so that users can get and share information easily.
- **Cost Benefits**
Storing information in one centralized database can also help to reduce costs and drive efficiency.

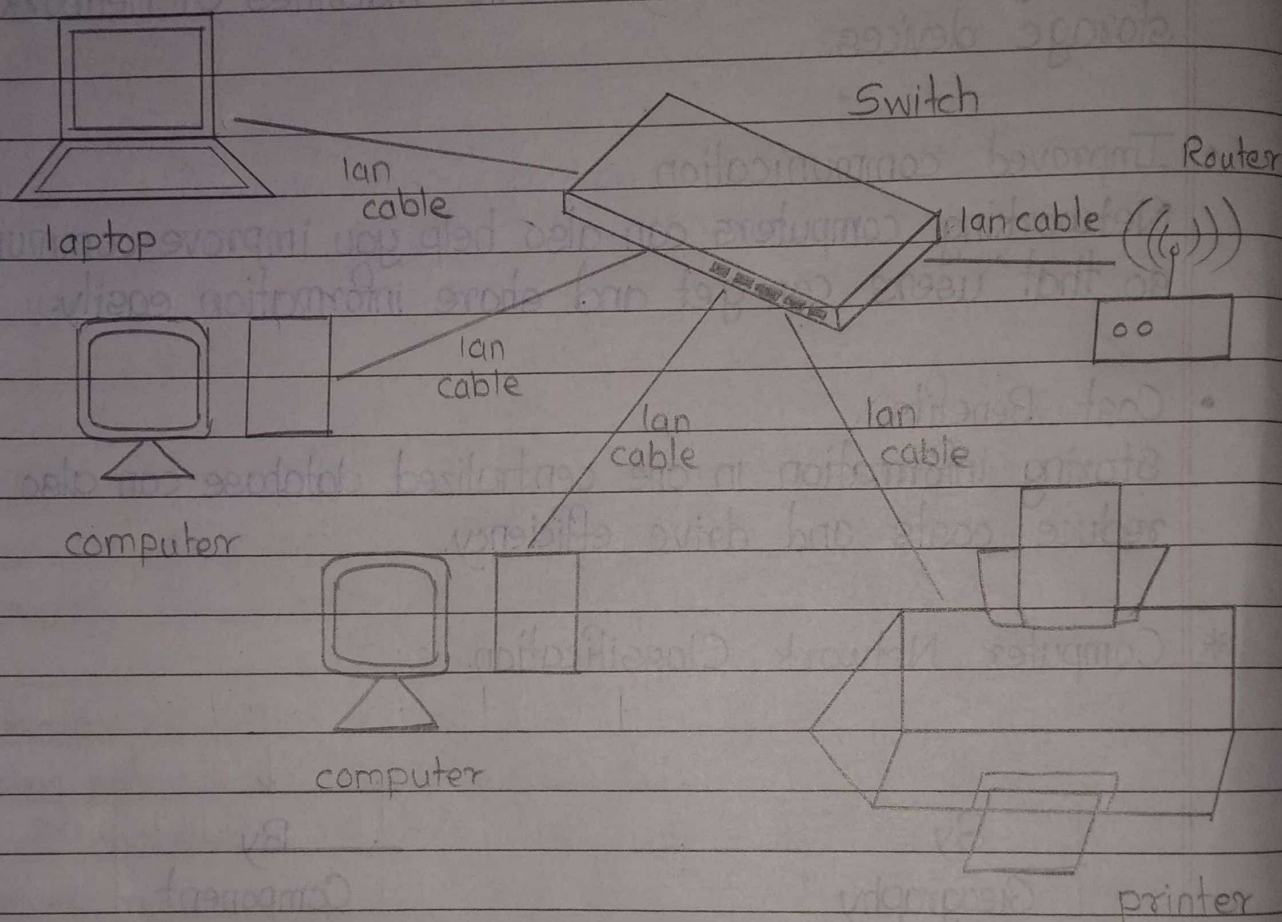
* Computer Network Classification



1. **Local Area Network (LAN)**
 - Most frequently used network.
 - It connects computers through a common communication path,

contained within a limited area, that is locally.

- It contains two or more computers connected over a server.
- Two important technologies involved in this network are Ethernet and Wi-fi.
- Ex. networking in home, school, library, college, office etc.

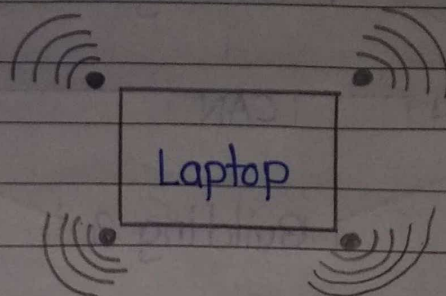


2. Personal Area Network (PAN)

- Most basic type of computer network.
- This network is restrained to a single person, that is, communication between the computer devices is centred only to an individual's work space.
- PAN offers network range of 10 metres.
- From a person to the device providing communication.
- Ex. USB, computer, phone, tablet, printer etc.

PDA

Smartphone

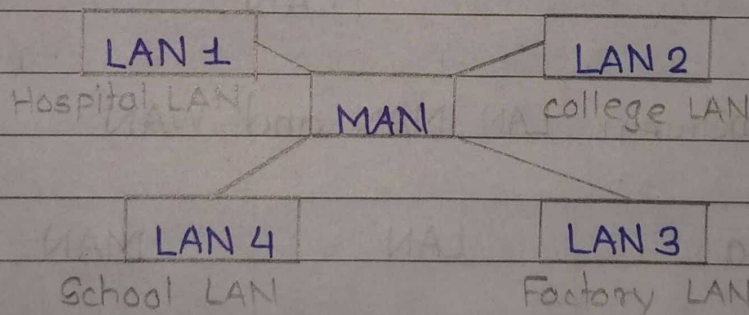


Tablet

Printer

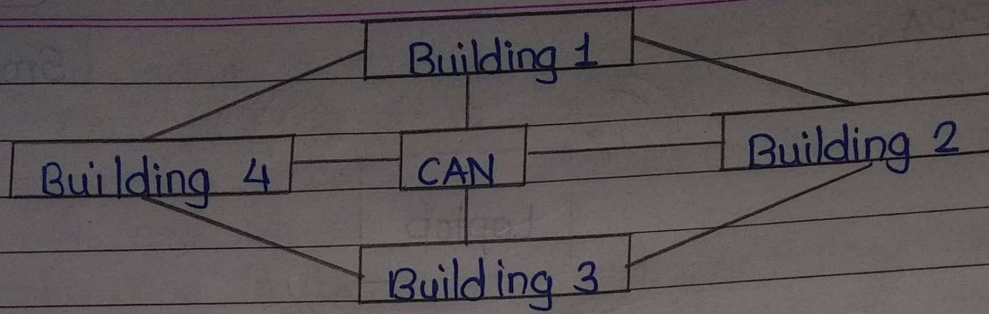
3. Metropolitan Area Network (MAN)

- MAN is larger than LAN but smaller than WAN.
- It connects computers over a geographical distance through a shared communication path over a city, town or metropolitan area.
- Ex. networking in towns, cities, a single large city, large area within multiple buildings etc.



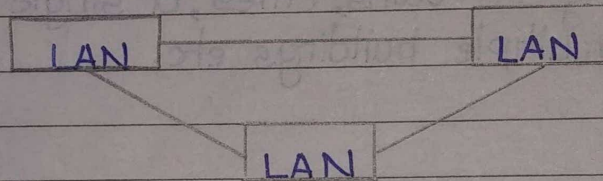
4. Campus Area Network (CAN)

- CAN is bigger than LAN but smaller than MAN.
- Usually used in places like a school or college.
- It covers limited geographical area that is, it spreads across several buildings within the campus.
- ex. network that ~~across~~ covers school, colleges, building etc.



5. Wide Area Network (WAN)

- It connects computers over a large geographical distance through a shared communication path.
- It extends over many locations
- WAN is group of local area networks that communicate with each other.
- ex - Internet



* Difference between LAN, MAN and WAN

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Criteria	LAN	MAN	WAN
Cost	low	high	higher
Network size	small	larger	largest
Speed	fastest	slower	slowest
Transmission media type	Twisted pair	Twisted pair & fibre optic cables	Fibre optic, radio wave & satellite
No. of computers	smallest	large	largest

* Classification of Network by their Component

i) Peer to peer network

ii) Client server based network

i) Peer to Peer Network

- A peer to peer network is analogous to a company that uses decentralized management where decisions are made locally.
- It is a network with all nodes acting as both servers and clients.
- A PC can access files located on another PC & can also provide files to other PC.
- All computers in peer to peer network has equal responsibilities and capabilities to use the resources available on network.
- It is useful for small network, when each computer maintains its own account & their security settings.
- Windows 2000 is Peer to Peer OS.
- Peer to Peer network don't have central control system thus it doesn't have central login process so it has very less security.

Features:

1. Useful for small networks with less than 10 computers.
2. Every computer can work as a client & server.
3. No central control system.
4. OS used is Windows 2000.
5. No additional software required.
6. Does not offer high network security.

Advantages:

1. Use less expensive computer hardware
2. Easy to administer.
3. No Network Operating System required.
4. More built-in redundancy.

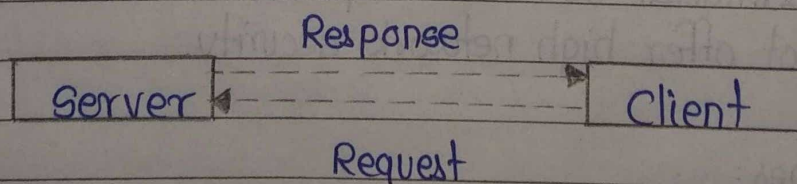
5. Easy setup
6. Lower cost

Disadvantages:

1. Individual performance is affected
2. Less security
3. Backup is difficult
4. Hard to maintain version control
5. No centralized management
6. Additional load on computer.

ii) Client Server Based Network

- A client is a computer running a program that requests services from a server.
- A server is simply a computer that makes the network resources available and provides service to other computers when they request for it.
- We can use any common OS such as Windows 7 for a client.
- But the server is loaded with a special OS such as Microsoft Windows Server 2008.
- The client places a request on the server machine when he wants an access to the centralized resources.
- The server responds to this request and sends the signal accordingly to the client.



Advantages:

1. The network is secure.
2. Better performance

3. Centralized backup
4. Higher reliability

Disadvantages:

1. Professional administration is required.
2. Consumes lots of memory and disk space.
3. Requires special network
4. Expensive dedicated hardware needs to be used.